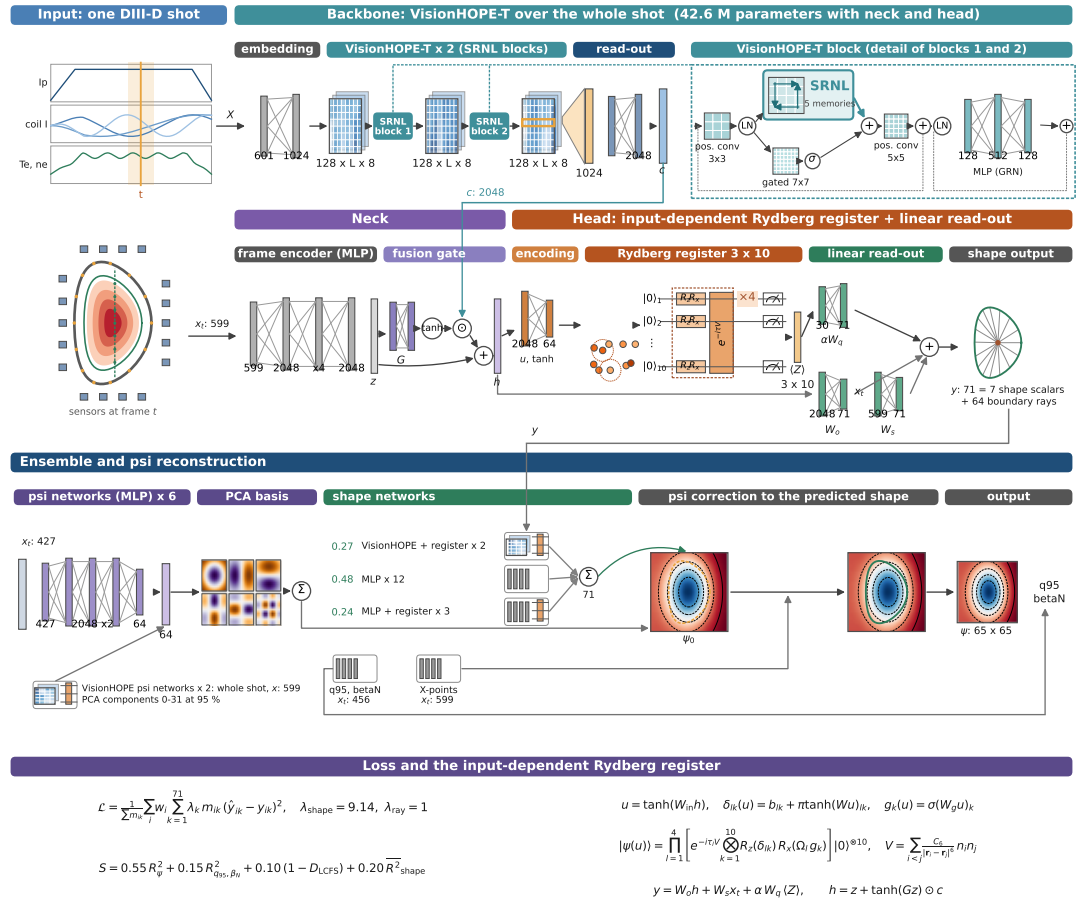


$$\mathcal{L} = \frac{1}{\sum m_{ik}} \sum_i w_i \sum_{k=1}^{71} \lambda_k \, m_{ik} \, (\hat{y}_{ik} - y_{ik})^2, \quad \lambda_{\text{shape}} = 9.14, \ \lambda_{\text{ray}} = 1$$

$$S = 0.55\, R^2_{\psi} + 0.15\, R^2_{q_{95},\,\beta_N} + 0.10\,(1 - D_{\text{LCFS}}) + 0.20\,\overline{R^2}_{\text{shape}}$$

$$u = \tanh(W_{in} h), \quad \delta_{ik}(u) = b_{ik} + \pi \tanh(W u)_{ik}, \quad g_k(u) = \sigma(W_g u)_k$$

$$|\psi(u)\rangle = \prod_{l=1}^{4} \left[e^{-i\tau_l V} \bigotimes_{k=1}^{10} R_z(\delta_{lk})\, R_x(\Omega_l g_k) \right] |0\rangle^{\otimes 10}, \quad V = \sum_{i<j} \frac{c_6}{|\mathbf{r}_i - \mathbf{r}_j|^6} n_i n_j$$

$$y = W_o h + W_s x_t + \alpha\, W_q\, \langle Z \rangle, \qquad h = z + \tanh(Gz) \odot c$$